

Investigating the Significance of Feature Extraction Methods for Audio Classification

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1. Introduction

Environmental Sound Classification (ESC) identifies non-speech sounds such as glass breaking or rain from audio recordings, with applications in healthcare monitoring, smart home systems, and criminal investigations. A crucial challenge in ESC is determining optimal audio representation strategies, as different feature extraction methods capture distinct acoustic properties.

Our research question investigates which audio tokenization strategy most effectively balances classification performance and computational efficiency. We compare: (1) handcrafted pooled statistics, (2) vision-based spectrogram models repurposed from computer vision, (3) audio-specific pretrained embeddings (VGGish). We also propose (4) a novel hybrid approach fusing VGGish embeddings with pooled statistics. We use FreeSound (N=51,197 samples, 74 classes, multi-label) as our primary dataset and validate generalization on ESC-50 (2,000 samples, 50 classes, 5 official folds).

2. Literature Review

A dominant paradigm treats spectrograms as images. Mushtaq et al. (2021) applied CNN's to ESC tasks, achieving very high accuracy on ESC-50. Vision Transformers (ViTs) show promise for images (Dosovitskiy et al., 2020), but their transferability to audio spectrograms remains underexplored. Jahangir et al. (2023) demonstrated that fusing transformers with CNNs yields better performance than single approaches on ESC-50.

Audio-specific models offer alternatives. VGGish (Hershey et al., 2017), pretrained on YouTube-8M's 70 million videos, provides embeddings designed for audio understanding. Fonseca et al. (2019) achieved 0.542 LWLRAP on FreeSound using CNNs, establishing a multi-label baseline.

Our work addresses gaps in systematic comparison of audio-specific versus vision-based pretraining and hybrid feature fusion approaches.

3. Methodology

Dataset and Task Definition

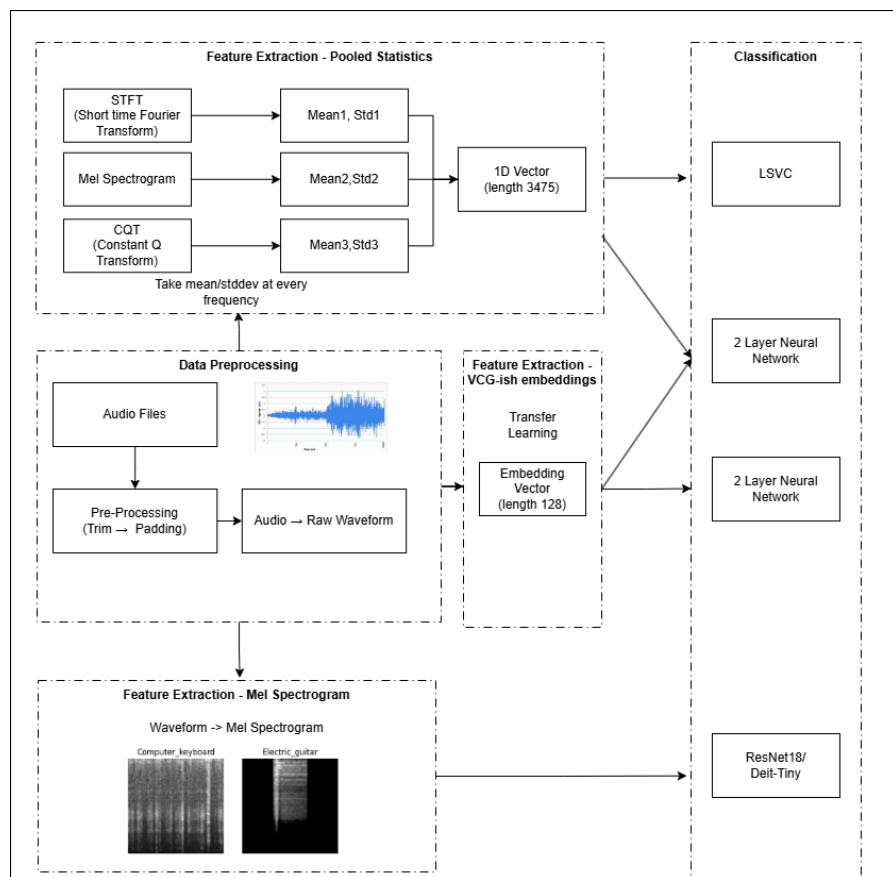
We use a subset of the FreeSound Dataset from DCASE 2019 (74 classes) and validate on ESC-50 (50 classes). To enable comparison with existing single-label ESC literature, we select the primary label per sample and evaluate using accuracy metrics.

Preprocessing

All files are resampled to 44.1 kHz mono, trimmed of leading/trailing silence, and zero-padded to 2 seconds. We apply loudness normalization to -20 LUFS.

Feature Extraction Methods (Figure 1) :

1. **Pooled statistics (handcrafted):** For each audio clip, we compute mean



and standard deviation across time for three spectral representations: STFT, mel-spectrogram, and CQT (bins 0–83). Yielding a 3,474-dimensional feature vector per sample

2. **VGGish embeddings (transfer learning):** We extract frame-level embeddings from the pretrained VGGish model (trained on YouTube-8M) and perform temporal pooling. These 128-dimensional embeddings are then fine-tuned end-to-end with a learned projection head.
3. **Mel-spectrogram images (vision models):** We convert audio to 128×128 log-mel spectrograms ($n_{\text{fft}}=2,560$, $\text{hop}=694$, 20 Hz–22.05 kHz range). These single-channel "images" are z-normalized per clip and fed to vision architectures pretrained on ImageNet. Light augmentation includes random time crops and Gaussian noise ($\sigma=0.02$).

Classifiers

We evaluate four distinct classification approaches. Linear Support Vector Classifier serves as our simplest baseline, operating directly on the 3,474-dimensional pooled statistics using hinge loss with L2 regularization controlled by hyperparameter C.

For vision-based approaches, we test two architectures on mel-spectrograms. MobileNetV4 serves as our CNN baseline. This model represents the dominant paradigm in audio classification where CNNs extract hierarchical features from spectrograms. DeiT-Tiny (Touvron et al., 2021) represents the Vision Transformer family, using 16×16 patches with distillation tokens and extensive regularization including dropout, stochastic depth, and label smoothing.

VGGish with fine-tuning (Hershey et al., 2017) provides our audio-specific baseline. We use the pretrained YouTube-8M model as a frozen feature extractor, feeding the 128-dimensional embeddings into a two-layer neural network classification head that we train end-to-end.

Finally, we propose a hybrid approach combining VGGish embeddings with pooled statistics. The 128-dimensional VGGish features are concatenated with the 3,474-dimensional pooled statistics, creating a 3,602-dimensional representation that is fed to a classification head. This fusion strategy aims to capture both learned semantic features and explicit acoustic measurements.

Hyperparameter Tuning and Nested Cross-Validation

We adopt a Nested Cross-Validation (NCV) strategy with k-folds methods avoiding overfitting and bias. 10-fold CV is used for the outer layer, while the inner layer uses 5-fold CV, all using 42 as the seed.

For each model, we select different hyperparameters to their different architecture. For Linear SVC, we choose C as 0.1, 1.0, 10.0, with l1, l2 penalty. For neural networks like MobileNetV4, VGGish and DeiT-Tiny, we fine tune with a learning rate of 5e-5, 5e-4, 1e-4 with different optimizers like Adam, SGD, Amsgrad. We use F1-Score to calculate performance on each class of multi-class classification f1 score on held-out test sets, and lwrap for overall performance.

4. Results

Due to most studies on ESC being done on single-label datasets, testing was carried out on ESC-50 for easier comparison with existing studies

Model Performance Comparison

Figure 2 presents classification accuracy across all four approaches on the FreeSound test set.

VGGish+Tabular performed the best at 89% accuracy, substantially outperforming MobileNetv4 and LinearSVC.

All models exceed the majority class baseline of 1.76%, with paired t-tests across 74 classes confirming significant improvements over DeiT-Tiny.

The Impact of Hybrid Features

Adding pooled statistics to VGGish embeddings yields a 25.18 percentage point improvement over VGGish standalone ($p=1.72e-11$, Cohen's $d=0.925$). This highly significant effect indicates 82% of classes (61/74) improved with hybrid features, while only 5% (4/74) worsened. Notably, 52.7% of classes improved by 10% or more, and 46.0% improved by 20% or more. The improvements were particularly dramatic for simple acoustic events: Squeak, Tap , Slam and Yell. These classes benefited from explicit spectral statistics that captured their distinctive acoustic ranges.

DeiT-Tiny as an Unfitted Baseline

DeiT-Tiny's 19.62% accuracy with 38 failed classes effectively serves as an unfitted baseline, demonstrating the difficulty of the task. Since ResNet-18 (11.7M parameters) achieved strong performance with similar parameters than DeiT-Tiny (5.7M), the failure is likely not parameter-related. Multiple architectural factors likely contribute, including patch-based processing that may destroy time-frequency structure of the spectrogram and training instability common with Vision Transformers on limited data. Given these challenges, we selected MobileNetV4 as our primary vision model comparison.

Figure 2: Classification Performance on FreeSound Dataset (74 classes)

Model	Accuracy (%)	Perfect Classes (100%)	Failed Classes (0%)	Model Type
VGGish + Tabular	89.24 ± 14.53	37	5	Hybrid
MobileNetV4	75.55 ± 20.44	14	2	Vision (CNN)
LinearSVC	57.44 ± 23.70	2	6	Classical ML
DeiT-Tiny	19.62 ± 17.47	0	38	Vision (Transformer)
Majority Class Baseline	1.76	0	73	Baseline

Note: Accuracy reported as mean ± standard deviation across 74 classes. Perfect Classes: 100% accuracy; Failed Classes: 0% accuracy. Bold indicates best performance.

Per-Class Performance Patterns

High-performing classes include stationary broadband sounds (traffic noise, rain, vacuum cleaner with >90% accuracy) and mechanical sounds with repetitive structure. Low-performing classes include human vocalizations showing inter-confusion due to shared acoustic properties and transient events where VGGish's temporal pooling loses onset information. VGGish+Tabular's 5 remaining failed classes (Cutlery_and_silverware, Drip, Raindrop, Acoustic_guitar, Crowd) all suffer from extremely low sample counts, demonstrating a practical minimum data threshold. However, the model achieves >80% accuracy on many rare classes, indicating general robustness to data scarcity.

Discussion

Feature Fusion for Learned and Handcrafted Features

The 25.18 percentage point improvement from combining VGGish embeddings with pooled statistics demonstrates that learned and handcrafted representations capture fundamentally complementary information. VGGish provides high-level semantic understanding from YouTube-8M pretraining, while pooled statistics (STFT/mel/CQT mean and std) explicitly encode low-level spectral details (precise frequency distributions and harmonic structure) that may be abstracted away in VGGish's deep layers.

The systematic nature of this improvement (82% of classes benefit) suggests the synergy is not class-specific but a general trend, where deep learned features may lose fine-grained acoustic details, while handcrafted features preserve explicit physical measurements. This is obvious in simple acoustic events (Squeak, Tap, Slam) where distinctive spectral signatures are captured by pooled statistics but not by VGGish embeddings.

Audio-Specific vs. Vision Pretraining

MobileNetV4's 75.55% accuracy demonstrates CNN effectiveness, yet it underperforms VGGish+Tabular by 13.69 percentage points. This gap highlights the value of audio-specific pretraining: YouTube-8M provides domain-matched representations that transfer more effectively than ImageNet's vision features. The finding suggests that for audio classification via spectrograms, practitioners should prioritize audio-pretrained models over repurposing vision architectures when possible.

We compared audio tokenization strategies for environmental sound classification, evaluating handcrafted pooled statistics, vision-based models, and audio-specific embeddings. The hybrid approach combining VGGish embeddings with pooled statistics achieved 89.24% accuracy, demonstrating a 25.18 percentage point improvement over VGGish alone. This substantial gain reveals that learned and handcrafted representations capture complementary information: semantic understanding from pretrained embeddings and explicit spectral details from pooled statistics. Despite testing both pretrained and non-pretrained configurations with various learning rates, Vision Transformers failed (19.62%), suggesting challenges in applying patch-based architectures to spectrograms in data-limited settings. For practitioners, we conclude an ensemble method using pretrained audio embeddings and pooled statistics offers an efficient use of limited data and time.

6. Conclusion and Future Work

Our findings address the research question of which tokenization strategy balances performance and efficiency. The hybrid approach combining VGGish embeddings with pooled statistics is the clear winner, demonstrating that learned and handcrafted representations remain powerfully complementary. For practitioners, this suggests feature concatenation rather than choosing between approaches. The computational overhead of computing both is offset by dramatic accuracy gains affecting 82% of classes. LinearSVC offers a fast alternative for resource-constrained scenarios, while MobileNetV4 provides a middle ground when audio-specific pretraining is unavailable.

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